

MD. MIFTAHUR RAHMAN SWAPNIL

Product Designer & Builder | AI, Frontend & Automation

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SUMMARY

Product Designer & Builder with an innovative and curious mindset, always exploring new ideas, technologies, and ways to turn them into working products. I enjoy building and experimenting with new things, whether it is to solve a real-world problem, learn something new, or gain hands-on experience. I use AI as a practical part of my workflow to simplify tasks, improve user experiences, and reduce time spent on repetitive work. Currently, I'm especially interested in AI automation and building smarter workflows that make everyday processes easier and more efficient.

SKILLS

- Product & Design:** UI/UX Design, Digital Product Design, Wireframing, High-Fidelity Prototyping (Figma), Design Systems, User Research
- Frontend Engineering:** Next.js 14/15 (App Router), React, TypeScript, Tailwind CSS, JavaScript (ES6+), HTML5/CSS3, Responsive UI
- Backend & Database:** Supabase (PostgreSQL, Row Level Security, Auth), Node.js, REST APIs, Google Drive API, Firebase
- AI & Research:** Knowledge Graphs & RAG Systems, Multi-LLM Orchestration, Relational GCN, Neo4j, PyTorch, Deep Learning for EEG, Python
- Tools & Automation:** Git & GitHub, n8n Automation, Telegram Bot API, Vercel, Docker (Basic), VS Code, LaTeX, Overleaf

FEATURED PROJECTS

- Personal AI Assistant — Mikasa** — Node.js, Telegram Bot API, Supabase, n8n, Multi-LLM Orchestration, Local PC Bridge **Active**
- Architected an autonomous personal AI operating system featuring persistent memory extraction, continuous learning, and multi-model failover (Gemini 2.5 Flash & OpenRouter).
 - Built bidirectional Telegram bot controls, cross-platform cron automation, and secure local PC bridges for terminal and desktop execution.
- Edu51Portal** — Next.js 14, TypeScript, Supabase PostgreSQL, Google Drive API, Tailwind CSS **Live**
- Designed and built an academic platform serving **~100 active BUBT students**, delivering instant routine lookups, notices, and course materials.
 - Integrated Google Drive API with Supabase to distribute 1,000+ academic PDFs securely with **\$0 cloud storage infrastructure overhead**.
- MuteBD** — Flutter, Dart, Google Maps API, Background Geofencing, Firebase **Completed**
- Engineered a location-based mobile application that automatically silences smartphones in designated quiet zones (mosques, hospitals, libraries) using real-time GPS geofencing.
- PawfectMatch** — React Native, Node.js, Firebase Realtime DB, AI Assistant **Completed**
- Designed and developed a pet adoption mobile platform connecting verified rescuers with adopters, featuring AI-guided pet care recommendations and matchmaking.

RESEARCH & PUBLICATIONS

- Relation-Aware Graph Retrieval over a Curriculum Knowledge Graph for Prerequisite QA** **Accepted (IEEE OMLET 2026)**
- IEEE OMLET 2026** (Nairobi, Kenya) — Accepted with Minor Revision (Paper ID: 1017, Scheduled for IEEE Xplore & Scopus).
 - Authors: Md. Jahidul Kamal Islam, **Md. Miftahur Rahman Swapnil**, Md. Asif Ali, Shrabani Das, Shefayatuj Johara Chowdhury (BUBT CSE).
 - Engineered a curriculum knowledge graph RAG system with a 2-layer R-GCN encoder and DistMult decoder over **418 nodes and 558 typed edges** in Neo4j, achieving **45.5% exact-set match** and 100% correct abstention rate.
- AI-Enabled Smart Classroom Monitoring and Safety Automation** **Completed (BUBT CSE)**
- Engineered an IoT smart classroom safety and environmental monitoring system integrating sensor telemetry (DHT11, PIR, Ultrasonic, Sound) with automated door actuators and alert buzzers.
- EEG-Based Motor Imagery Classification** **Ongoing / Running (2026)**
- Investigating deep learning architectures for assistive neuro-rehabilitation and paralysis motor control interfaces.

EDUCATION

- Bachelor of Science in Computer Science & Engineering (CSE)** **CGPA: 3.60 / 4.00**
Bangladesh University of Business and Technology (BUBT) | Intake 51 **2022 – 2026 (Expected)**
- Higher Secondary Certificate (HSC) — Science** **GPA: 5.00 / 5.00**
Shaheed Police Smrity College **2021**
- Secondary School Certificate (SSC) — Science** **GPA: 5.00 / 5.00**
Kadirabad BL High School, Pirganj, Rangpur **2019**

LEADERSHIP & COMMUNITY

- BASIS Students' Forum — BUBT Chapter** — Member, Graphics Designer & Media and Publication Secretary **2023 – 2026**
- Directed creative visual branding, digital promotional media, and publication outreach for tech seminars, workshops, and industry networking events.
- BUBT IT Club** — General Member **2022 – Present**
- Internal University Event Management** — Active Coordinator & Event Organizer **2022 – Present**

LANGUAGES

- Bangla** — Native / First Language
- English** — Professional Working Proficiency